Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Built **evaluation harnesses** scoring generative output quality, safety, **hallucination rates**, and grounding across model and agent versions, gating releases on **regression suites** and **A/B tests**.
- Trained and fine-tuned **PyTorch** models on **multi-GPU clusters** (AWS, GCP, CoreWeave) with **NVIDIA/CUDA**, using **quantization** for efficient inference.
- Built observability and **autoscaling** for **vLLM** inference serving with **Prometheus/Grafana**, and kept distributed training on **Slurm** running through node loss.
- Led a production **multi-agent GenAI system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** across GPT, Claude, Llama, and Gemini, handling **millions of daily transactions**.

QureAI | *AI Engineer* Mar 2026 – Present

- Built **multi-modal computer vision** models on **CNN and ResNet** backbones with **UNet segmentation** for medical image understanding in **PyTorch**.
- Extended **vLLM-class** serving with **CUDA/Triton** kernels, distributed **KV cache**, and **speculative decoding**, alongside **TensorRT-LLM**.
- Built **PySpark** jobs for large-scale dataset preparation feeding model training and **evaluation pipelines**, with **Langfuse** tracing on production agents.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Improved vision model accuracy **9.6 points** and cut false negatives **31%** by systematically evaluating **CNN, ResNet, UNet, and ViT** architectures, augmentation, and decision thresholds.
- Raised inference throughput **2.4x** and cut **GPU memory 27%** by profiling and optimizing batching, preprocessing, tensor movement, and concurrency in **Python/C++** services.
- Cut model-related production regressions **28%** and validation time from **6 hours to 90 minutes** with an automated **model-evaluation and deployment-gating** framework.
- Raised ML service availability to **99.9%** and cut **p99 latency 32%** with **Python/C++** serving infrastructure, and led **design and code reviews**.

Projects

HoG-ViT (M.S. Thesis) `github.com/ganeshhgupta/HoG-ViT`

- **Vision Transformer** that folds **Histogram of Oriented Gradients** features into **patch embeddings** for image classification, built and evaluated in **PyTorch**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- **LLM observability** platform that traces model calls, latency, and token usage across versions, with a **FastAPI** backend and **React/TypeScript** dashboard.

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that triages production alerts, routes each to a specialized sub-agent with **tool use**, and writes a structured **root-cause brief**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Python, C++, Java, SQL, TypeScript

GenAI and Vision: PyTorch, Vision Transformers, CNN, ResNet, UNet, LLMs, Multi-modal Models, RAG, Fine-tuning, Quantization

ML Infrastructure: vLLM, TensorRT-LLM, CUDA, Triton, Multi-GPU Training, Slurm, Kubernetes, PySpark, GCP, AWS

Evaluation: Model Evals, Regression Suites, Release Gating, A/B Testing, Hallucination Detection, Langfuse

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer